

Quantum Mechanics

Haoran Zhou^{1,2}

¹ Institute of Theoretical Physics,
Chinese Academy of Sciences, Beijing 100049, China

² School of Physical Sciences,
University of Chinese Academy of Sciences, Beijing 100049, China

Preface

This is my notes on Quantum Mechanics. The main reference is *Modern Quantum Mechanics* by J.J. Sakurai, *Mathematical Methods in Quantum Mechanics With Applications to Schrödinger Operators* by Gerald Teschl and *Quantum Theory for Mathematicians* by Brian C. Hall. Other references include *Quantum Mechanics* by Cohen-Tannoudji and *Quantum Theory, Group Theory, and Representations* by Peter Woit.

Contents

1	Quantum Mechanics of A Single Spinless Particle	6
1.1	Basic Concepts	6
1.2	Canonical Quantization	6
1.2.1	Wavefunction	6
1.2.2	Observable	6
1.2.3	Schrödinger Equation	6
1.3	Schrodinger/Heisenberg/Interaction Picture	6
1.4	Uncertainty Principle	6
2	Exact Solutions	7
2.1	Free Particle	7
2.2	Infinite Potential Wall and Finite Potential Wall	7
2.3	Polynomial Potential	7
2.4	Harmonic Oscillator	7
2.5	Hydrogen-like Atom	7
2.6	Periodic Potential	7
3	Quantum Mechanics of A Single Particle with Spin	8
3.1	Spin wavefunctions	8
3.2	Galilean Group	8
3.3	Representations of Galilean Group	8
3.4	Spin and Angular Momentum	8
3.5	Addition of Angular Momentum	8
4	Quantum Mechanics of Many-body Systems	9
4.1	Wavefunctions of Many-body Systems	9
4.2	Identical Particles	9
5	Quantum Properties	10
5.1	Randomness	10
5.2	Decoherence Theory	10
5.3	Non-locality	10

5.4	Contextuality	10
6	Symmetry	11
6.1	Symmetry and Conservation	11
6.2	Symmetry and Degeneracy	11
6.3	Symmetry of Harmonic Oscillator and Hydrogen-like Atom	11
6.4	Parity Reversal	11
6.5	Time Reversal	11
7	Density Operator Formalism	12
7.1	Density Operator	12
7.2	Reduced Density Operator	12
7.3	Wigner Function	12
8	Path Integral Formalism	13
8.1	Path Integral	13
8.2	Wiener Measure	13
8.3	Feynman-Kac Formula	13
9	Approximation Methods	14
9.1	Time-Independent Perturbation Theory	14
9.2	Time-Dependent Perturbation Theory	14
9.3	Variational Method	14
9.4	WKB Approximation	14
9.5	Adiabatic Approximation	14
10	Scattering Theory	15
10.1	In-State and Out-State	15
10.2	Scattering Amplitude	15
11	Atoms	16
11.1	Hydrogen-like Atom	16
11.2	Fine Structure of Hydrogen-like Atom	16
11.3	Zeeman Effect	16
11.4	Multi-electron Atoms	16
12	Quantum Entanglement	17
12.1	Entanglement Entropy	17
12.2	Effects of Entanglement	17
12.3	Bipartite Entanglement	17
12.4	Multipartite Entanglement	17

13 Relativistic Quantum Mechanics	18
13.1 Klein-Gordon Equation	18
13.2 Dirac Equation	18

Chapter 1

Quantum Mechanics of A Single Spinless Particle

1.1 Basic Concepts

1.2 Canonical Quantization

1.2.1 Wavefunction

1.2.2 Observable

1.2.3 Schrödinger Equation

1.3 Schrodinger/Heisenberg/Interaction Picture

1.4 Uncertainty Principle

Chapter 2

Exact Solutions

2.1 Free Particle

2.2 Infinite Potential Wall and Finite Potential Wall

2.3 Polynomial Potential

2.4 Harmonic Oscillator

2.5 Hydrogen-like Atom

2.6 Periodic Potential

Chapter 3

Quantum Mechanics of A Single Particle with Spin

3.1 Spin wavefunctions

3.2 Galilean Group

3.3 Representations of Galilean Group

3.4 Spin and Angular Momentum

3.5 Addition of Angular Momentum

Chapter 4

Quantum Mechanics of Many-body Systems

4.1 Wavefuctions of Many-body Systems

4.2 Identical Particles

Chapter 5

Quantum Properties

5.1 Randomness

5.2 Decoherence Theory

5.3 Non-locality

5.4 Contextuality

Chapter 6

Symmetry

6.1 Symmetry and Conservation

6.2 Symmetry and Degeneracy

6.3 Symmetry of Harmonic Oscillator and Hydrogen-like Atom

6.4 Parity Reversal

6.5 Time Reversal

Chapter 7

Density Operator Formalism

7.1 Density Operator

7.2 Reduced Density Operator

7.3 Wigner Function

Chapter 8

Path Integral Formalism

8.1 Path Integral

8.2 Wiener Measure

8.3 Feynman-Kac Formula

Chapter 9

Approximation Methods

9.1 Time-Independent Perturbation Theory

9.2 Time-Dependent Perturbation Theory

9.3 Variational Method

9.4 WKB Approximation

9.5 Adiabatic Approximation

Chapter 10

Scattering Theory

10.1 In-State and Out-State

10.2 Scattering Amplitude

Chapter 11

Atoms

11.1 Hydrogen-like Atom

11.2 Fine Structure of Hydrogen-like Atom

11.3 Zeeman Effect

11.4 Multi-electron Atoms

Chapter 12

Quantum Entanglement

12.1 Entanglement Entropy

12.2 Effects of Entanglement

12.3 Bipartite Entanglement

12.4 Multipartite Entanglement

Chapter 13

Relativistic Quantum Mechanics

13.1 Klein-Gordon Equation

13.2 Dirac Equation